

Detecting Emotions using EEG and Deep Learning

Student Name: Madeline Boyes

Supervisor Name: Gokberk Kabacaoglu

Submitted as part of the degree of BSc Computer Science to the
Board of Examiners in the Department of Computer Sciences, Durham University

Abstract—This dissertation presents a deep learning pipeline for classifying emotions from electroencephalogram (EEG) signals, with applications in clinical rehabilitation. Recognising emotions accurately is vital for optimising patient recovery post-stroke or brain injury. Traditional emotion recognition methods, such as facial analysis or self-reports, are often unsuitable in such contexts. To address this, the project employs EEG data due to its objective measurement of brain activity. A novel artifact removal pipeline using Variational Mode Decomposition with Canonical Correlation Analysis (VMD-CCA) is implemented to mitigate motion-induced noise. Emotion classification is achieved using a custom Convolutional Neural Network (CNN) trained on nonlinear EEG features such as entropy and fractal measures. The model achieved a classification accuracy of 91.5%, demonstrating robustness across six emotion classes using the SEED-VII dataset. The system is modular, generalisable, and well-suited for future real-time deployment in rehabilitation settings. This work provides a promising foundation for automated emotion-aware rehabilitation support systems, overcoming key limitations in existing EEG-based approaches.

Index Terms—Machine Learning, Medical Information, Modelling Human Emotion, Physiological Measures



1 INTRODUCTION

1.1 Background and Motivation

EMOTIONS result from the body's reactions to events and situations. According to Ekman [1], six basic emotions are innate: happiness, sadness, surprise, fear, anger, and disgust. The rest of the emotions are made up of combinations of these 6 basic emotions. Emotions influence almost every aspect of humans and our behaviour. They are integral to mental health and can also impact physical well-being [2]. Emotional states play a key role in medical recovery and rehabilitation. For instance, research has shown that positive emotions following a stroke can enhance rehabilitation outcomes [3], [4]. Seale [4] demonstrated that individuals who experienced a rise in positive emotions over a three-month span were significantly more likely to regain functional abilities. Conversely, Eslinger [5] demonstrated that negative emotions contributing to post-stroke depression can impede the recovery process. Therefore, it is extremely important to measure these emotions accurately.

The motivation behind this project is to enhance rehabilitation outcomes for individuals recovering from strokes and brain injuries by accurately measuring their emotional responses. By identifying which rehabilitation exercises elicit positive emotional engagement and which trigger negative emotional reactions, healthcare professionals can tailor recovery programs that support both emotional well-being and functional recovery.

Historical methods include surveys like the Self-Assessment Manikin (SAM) or the use of Artificial Intelligence (AI) with facial imaging. SAM is inadequate as it is dependent on the user's understanding and expression of their own emotions whereas, AI with facial imaging would fail when a patient has a stroke or a brain injury

LIST OF ACRONYMS

Acronym	Definition
AI	Artificial Intelligence
ADMM	Alternating Direction Method of Multipliers
BSS	Blind Source Separation
CCA	Canonical Correlation Analysis
CNN	Convolutional Neural Network
DWT	Discrete Wavelet Transform
DT-CWT	Dual-Tree Complex Wavelet Transform
EEG	Electroencephalography
EEMD	Ensemble Empirical Mode Decomposition
HE	Hurst Exponent
HFD	Higuchi Fractal Dimension
ICA	Independent Component Analysis
LSTM	Long Short-Term Memory
MC-LS-SVM	Multi-Class Least Squares Support Vector Machine
PCA	Principal Component Analysis
QTFD	Quadratic Time-Frequency Distribution
ReLU	Rectified Linear Unit
RCA	Reverse Correlation Analysis
SAM	Self-Assessment Manikin
SE	Sample Entropy
SNR	Signal-to-Noise Ratio
SVM	Support Vector Machine
TFR	Time-Frequency Representation
TE	Tsallis Entropy
VF	Volterra Filter
VMD	Variational Mode Decomposition
VMD-CCA	Variational Mode Decomposition with Canonical Correlation Analysis

which means that they may display emotions on their faces differently. Electroencephalography (EEG) scanners measure the person's innate brain signals making the classified emotion objective. Therefore, EEG scanners are the most reliable method for emotion classification in these patients. Advancing their application in clinical settings is crucial to improving emotion recognition and, ultimately, patient rehabilitation outcomes.

Emotions can be measured across two dimensions: valence (which indicates whether an emotion is positive or negative) and arousal (which reflects its intensity). In order to produce each emotion, the body experiences the stimulus which is processed by several brain regions. The brain processes emotions through three main regions. The hypothalamus regulates emotional responses by triggering physiological reactions such as increased heart rate and sweating. The amygdala evaluates external stimuli, assessing emotional significance and potential threats. Meanwhile, the hippocampus links emotions to past experiences, aiding in memory formation and emotional regulation. Since different emotions activate distinct brain regions, consistent emotional responses activate the same brain regions on an EEG scan. This principle forms the foundation of EEG-based emotion recognition.

EEG scanners work by detecting the brain's electrical activity using metal electrodes placed on the scalp [6]. Different frequency bands relate to different brain activity, these bands can be decomposed into: delta, theta, alpha, beta and gamma. These can be shown in Table 1.

TABLE 1
EEG Frequency Bands and Their Associated Brain States [6]

Band	Frequency (Hz)	Associated Brain State
Delta	1-4	Deep sleep, unconscious mind
Theta	4-7	Subconscious, sleep, dreaming
Alpha	8-15	Relaxed mental state, brain activation
Beta	16-31	Active mind, concentration
Gamma	31-60	Intense brain activity and cognition

Due to the delta and theta bands being linked to low level states of mind, emotions are only associated with alpha, beta and gamma bands. Additionally, signals with a greater frequency than gamma are unrelated to emotions and are called artifacts. Artifacts can be produced from electrical signals from surrounding medical equipment, or from any movement from the patient. This has meant that when measurements are taken, patients have to be completely still with eyes shut in order to take a valid measurement. This project aims to experiment with the methods from the literature to remove these artifacts so that measurements can be taken when patients are moving.

In order to classify emotions we used deep learning models, specifically Convolutional Neural Networks (CNN). The advantage of using CNN models is that they can capture spatial information from the EEG signals and can discover the hidden patterns in the data which link to each emotion. Additionally, deep learning models can learn from vast quantities of data. This is necessary as each measurement taken from patients contains 32-64 channels of electrodes.

1.2 Research Questions

This project aims to accurately measure patients' emotions during active rehabilitation to help healthcare professionals tailor rehabilitation plans. By identifying which exercises challenge patients without causing distress, we can optimise recovery. To achieve this, the project must reduce motion-induced artifacts while maintaining accurate emotion classification. The objectives of this project are:

- Implement pre-processing techniques for artifact and noise removal
- Implement feature extraction for deep learning model
- Implement CNN and LSTM model
- Fine-tune models to have a classification accuracy over 90%
- Produce a polished piece of software

Therefore, this project seeks to answer the research question: How can EEG-based emotion recognition be used to assess patient emotional levels during rehabilitation while reducing the impact of motion artifacts?

2 RELATED WORK

This section reviews key areas of research that inform the development of this project's EEG-based emotion recognition system. It is structured around the main pipeline of processing EEG data: reducing the number of channels to improve computational efficiency, removing movement artifacts to ensure signal quality, extracting meaningful features from noisy and non-linear signals, and applying deep learning models for accurate emotion classification. Each subsection highlights relevant approaches and findings in the literature, identifies their limitations, and establishes a foundation for the methods employed in this project, which are described later.

2.1 Channel Reduction

Channel reduction reduces the number of output channels from the EEG signals. This is necessary to reduce the data storage, processing and learning time with deep learning methods while maintaining the same level of accuracy. To successfully do this, the method must maintain the most important electrode channels.

There are two main approaches to reducing EEG channels: reducing the number of electrodes during measurement or removing channels after data collection. There are many studies to reduce the number of electrodes on the scalp. These methods improve the ergonomics of EEG devices making them easier to wear the EEG scanner on the head as it is less likely that electrodes will become unattached when the patient moves around. The second method of channel reduction is particularly relevant in studies utilizing pre-existing open-source EEG datasets, where data collection cannot be tailored to specific experimental needs.

One approach is filtering methods, Dura et al. [7] ranked electrodes based on scoring techniques, specifically reverse correlation (RCA). This method, initially yielded the top 4 electrodes for individuals. However, this approach did

not generalise well across individuals, so the number of channels was increased to allow for a single configuration that could be applied to any subject without requiring prior knowledge about them. In order to do this, Dura et al. constructed a shared set of electrodes across all subjects by removing any electrodes that were not included in at least one participant's top four. This outputted 11 channels to be used for any participant, the placement of these can be shown in Figure 1b. Using only these 11 channels resulted in just a 1% drop in accuracy but reduced training time by 58%.

Another approach is wrapper-based methods, which use learning algorithms to evaluate subsets of channels. One study [8] applied this to the DEAP dataset and used particle swarm optimisation and cuckoo search to find the 11 most dominant channels for emotion classification. These 11 channels were discovered using 31 participants and the results were compared with using the entire 32 channels available.

The final set of methods are embedded methods, like group sparse Canonical Correlation Analysis (CCA), select features while incorporating group influences. One study identified 4 channels with about 80% accuracy across three emotion classes [9]. Another used Graph Neural Networks (GNNs) to achieve 73% accuracy on the SEED dataset while also serving for channel reduction [10].

One of the main challenges in selecting an appropriate channel reduction method is the lack of validation across alternative datasets in many existing studies. This lack of cross-dataset testing suggests that some methods may not generalise well, with reported accuracies failing to replicate under different conditions. This is especially problematic in real-world applications—particularly in clinical or medical settings—where consistency and generalisability are crucial. If a method performs well only on a specific dataset but fails elsewhere, it could lead to unequal outcomes, with some patients receiving more accurate assessments than others. For this reason, the approach proposed by Dura et al. [7] is particularly valuable. Dura et al. identified 11 EEG channels that consistently contributed to accurate emotion classification across participants, offering a strong basis for generalisable feature reduction. After reducing the signals to the most informative channels, the next step involves addressing another major challenge in EEG analysis: the removal of movement artifacts to ensure the signals accurately reflect neural activity to classify emotions accurately.

2.2 Removing Movement Artifacts

To accurately detect emotions whilst patients are moving, it is crucial to effectively remove artifacts caused by muscle movements. However, excessive or improper artifact removal can also eliminate important brain signals, significantly degrading the accuracy of emotion classification. Therefore, it is essential for a good balance to be struck between removing the artifacts and preserving meaningful neural activity for reliable emotion recognition.

Several methods already exist for removing noise from EEG signals. However, according to Chen [11] 85% of studies do not involve removing muscle artifacts, as the muscle and brain signal frequencies often overlap. This causes

most basic filtering methods to fail. Therefore, we will only review source separation approaches.

Common noise elimination techniques include adaptive filtering (AF), principal component analysis (PCA), and singular value decomposition (SVD), each with specific limitations. For example, PCA struggles to distinguish artifacts from brain signals when their amplitudes are similar [12]. Volterra Filter (VF) theory, first explored by Wiener, effectively models non-linear systems [13]. However, its exponential growth in computational complexity makes it impractical for large-scale EEG data and deep learning applications, and thus unsuitable for this project.

We are focusing on Blind Source Separation (BSS) techniques due to their ability to function without prior knowledge of the subject's activity and minimal setup requirements for medical staff. For example, Independent Component Analysis (ICA) has been widely used to separate EEG signal components and remove artifacts [14]. Variants such as Variational Mode Decomposition (VMD-ICA) [15] and Enhanced Empirical Mode Decomposition (EEMD-ICA) [16] have been built on this foundation, but all suffer from high computational costs. To address this, Hossain proposed Variational Mode Decomposition with Canonical Correlation Analysis (VMD-CCA) [17]. This method decomposes signals into modes with distinct central frequencies and then evaluates their correlations. Although VMD-CCA has only been tested on single-channel data and not yet applied to emotion classification, it outperformed ICA-based methods—including VMD-ICA and EEMD-ICA—in both accuracy and computational efficiency when tested on real and simulated noisy signals [18]. It also proved more accurate than EEMD-CCA [19]. Therefore, VMD-CCA offers a balance between effectiveness and computational efficiency.

While many artifact removal methods show promise, most have been developed and evaluated using only single-channel data and have not been adapted for emotion classification tasks. To the best of my knowledge, this work is the first to apply VMD-CCA for artifact removal in multi-channel EEG data specifically for emotion classification. After artifact removal, the next step is feature extraction, which is essential for feeding structured input into deep learning models.

2.3 Feature Extraction

Feature extraction is crucial because EEG signals are inherently complex and non-linear. By extracting relevant features, AI systems can learn the meaningful features associated with each emotion.

Several studies have explored variations of wavelet transform for feature extraction, including discrete wavelet transform (DWT) [20], tunable Q wavelet transform [21] and dual-tree complex wavelet transform (DT-CWT) [22]. Wavelet transform decomposes the signal to analyse different frequencies components over time [23].

Another approach proposed in several studies involves converting EEG signals into Time-Frequency Representation (TFR) using techniques such as S-transform [25], quadratic time-frequency distribution (QTFD) [24]. These methods were only used to classify 4 emotions. Islam used channel correlation [26], which compares all combinations of

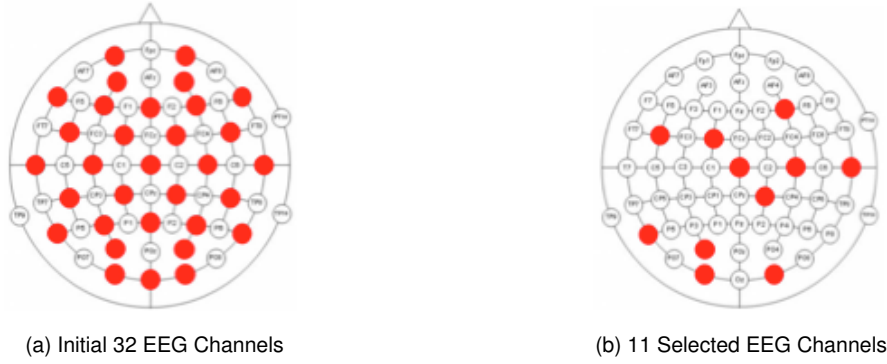


Fig. 1. The result of [7] channel reduction algorithm giving 11 selected channels for emotion recognition

the channels and produced a matrix image to input to a classifier. This only achieved around 75% accuracy for binary classification (valence/arousal), which highlights that this method may have limitations in distinguishing specific emotions.

A key limitation of the studies discussed above is the variation in the number of emotion classes used for classification, which makes it difficult to directly compare their results or assess their effectiveness in distinguishing between specific emotions.

Taran et al. [27] proposed a feature extraction method which computes Sample Entropy, Tsallis Entropy, Higuchi Fractal Dimension, and Hurst Exponent for each EEG channel. This method achieved 88% classification accuracy when using multi-class least squares support vector machine (MC-LS-SVM) classifier across four emotional states—happy, fear, sad, and relaxed. Building on this, the current work aims to enhance performance across the six basic emotion categories. These extracted features form the input to the deep learning models, enabling them to learn complex patterns associated with different emotional states.

2.4 Deep learning methods

Deep learning excels at handling large amounts of data provided the hyper-parameters are carefully selected. The key goal is to design models that are sufficiently general, ensuring they can be effectively applied to any subject.

2.4.1 Convolutional Neural Networks

CNNs are designed to preserve spatial information and learn from it through three main layers: convolutional, pooling, and fully-connected. The convolutional and pooling layers extract features, while the fully-connected layer performs classification [28]. Pooling reduces the representation size while retaining key features, with common methods like average, max, and global pooling. The activation function in the fully-connected layer adds non-linearity, improving the model's ability to capture complex patterns [29]. While Long Short Term Memory (LSTM) networks and hybrid models offer benefits in capturing temporal dependencies, this project focuses on CNNs due to their efficiency in feature extraction from EEG signals.

While existing methods have demonstrated promising results in isolation, there remains a lack of integrated systems that combine robust preprocessing, feature extraction,

and generalisable deep learning architectures for emotion recognition suitable for medical applications.

3 METHODOLOGY

The system takes 62 Channels of EEG data as input and outputs a classification of emotion. The pipeline for our system design is illustrated in Figure 2. Later, the individual components of the system are described in detail. Firstly, the raw EEG signals were preprocessed to remove baseline drift and artifacts using polynomial detrending, high-pass filtering, VMD-CCA. Secondly, a set of non-linear features—including Sample Entropy, Tsallis Entropy, Higuchi Fractal Dimension, and Hurst Exponent—was extracted from each EEG segment for every selected channel. These features captured the complex temporal and fractal properties of the EEG signal. The resulting feature vectors, were used as input to a custom-designed CNN for emotion classification. The methodology was designed to ensure both controlled validation (using synthetic artifacts) and realistic performance evaluation (using real-world emotional EEG data).

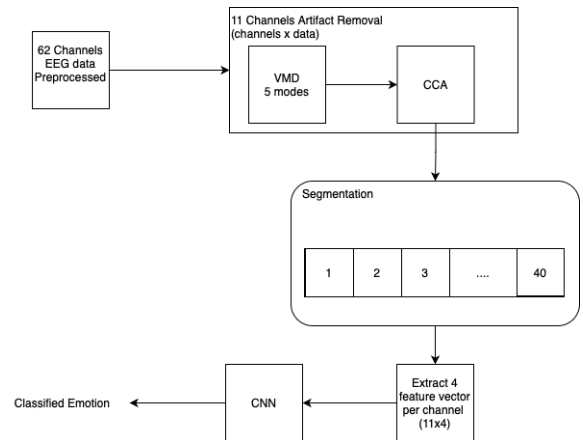


Fig. 2. Overview of the EEG emotion classification pipeline. The system takes raw EEG signals from 62 channels as input. Preprocessing steps include polynomial detrending, high-pass filtering, and artifact removal using Variational Mode Decomposition with Canonical Correlation Analysis (VMD-CCA). Following preprocessing, non-linear features are extracted from each segment of each channel. These features are then used as input to a custom CNN, which outputs the final emotion classification.

3.1 Datasets

In order to create and evaluate the methods, 2 datasets were used for the 2 main sections of the project: artifact removal and emotion classification. The first dataset is a motion artifact contaminated dataset [30] which was used to develop and evaluate the artifact removal methods. The second dataset was SEED-VII dataset [31] which was used to perform more realistic experiments using deep learning.

The motion artifact contaminated dataset was created specifically for validating artifact removal techniques. This dataset contains one pair of EEG signals 690 nm apart for 23 different trials. One signal is an untampered ground truth signal whereas, the 2nd signal had artificial movement added every 2 minutes. This means that when evaluating the artifact removal methods the outputted "clean" signal can be compared with the first untampered signal. An example of this signal can be shown in Figure 3.

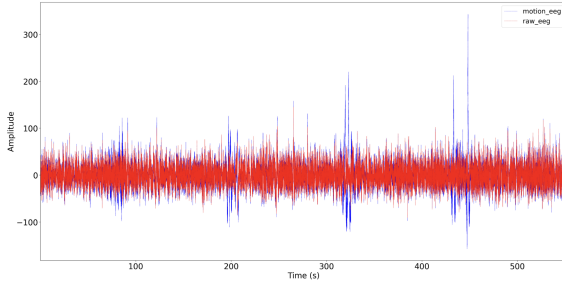


Fig. 3. 9 minutes of motion contaminated data. Red is the "ground truth" raw data, Blue is the artificially contaminated data

The SEED-VII dataset uses 6 basic emotion categories to label participant responses to stimuli. It was created with 20 participants with 80 trials each. The dataset consisted of 10 males and 10 females with an age range of 19 to 26 years. The 80 trials were separated over 4 sessions which were separated by over 24 hours each. Each trial was separated into watching an emotional video and self-assessment of the effects of that video. The data was recorded with 62 channels of EEG and produced a rating for each trial, which ranged from 0-1 (0 is no effect, 1 if emotion felt correlated with emotion intended to be induced). During the collection of data, participants eye movements were recorded. This facilitated as the artifacts in the dataset.

In this project, we chose to only process EEG data associated with ratings of ≥ 0.5 . The rationale behind this decision lies in the need to ensure that the data used for training the deep learning model accurately reflects the intended emotional response. Specifically, ratings below 0.5 were excluded because they likely represent instances where the stimulus did not effectively evoke the desired emotional response. Including such data, where the emotional effect is minimal or absent, could introduce noise into the training process. As a result, training the model on such data could degrade its performance by causing it to learn patterns that are not genuinely representative of the intended emotional states. By filtering out these lower-rated signals, we aimed to improve the quality of the data and, consequently, the performance of the model.

3.2 Pre-Processing and Artifact Removal Pipeline

The pre-processing and artifact removal pipeline was developed and tested using the motion artifact dataset. Once this proved to work well, the same process was then applied to the SEED-VII dataset to develop the emotion classification model.

To prepare the EEG data for classification, a structured pre-processing pipeline was implemented. This pipeline consisted of the following key steps:

- 1) Detrend Data
- 2) Highpass filtering
- 3) Extract 11 channels
- 4) Remove Artifacts using VMD-CCA

3.2.1 Standard Pre-processing

EEG signals frequently exhibit a gradual upward trend over time, with each channel displaying its own distinct drift pattern. This happens due to the skin contact with the EEG electrodes where the electrical signals build up over time due to sweat or gel on the skin. This means that the 'slow drift' can be stopped by piercing the skin, however, this is not a viable solution as we are interested in keeping the measurement as minimally invasive as possible. To correct this, robust detrending [32] was chosen over the classic linear detrending. This is because the signals which the project is interested in is artifact contaminated data which can alter how linear detrending works. This effect was demonstrated by De Cheveigné [32] where the presence of artifacts caused classical detrending methods to produce large deviations in the signal, resulting in distortions that deviated significantly from the true underlying trend.

Specifically, a baseline drift removal function was implemented using polynomial curve fitting. Given a one-dimensional signal $x(t)$, a polynomial of a specified order n (we used 3 in this work) was fitted to the signal using least squares estimation. The time vector t was generated as a sequence of indices corresponding to the length of the signal. To identify and remove the drift in the signal, a smooth curve was fitted to the data using polynomial regression. This involves finding a polynomial that closely follows the overall shape of the signal without capturing the fast, meaningful fluctuations. The final corrected signal was obtained by subtracting the fitted polynomial baseline from the original signal:

$$x_{corrected}(t) = x(t) - \hat{x}_{baseline}(t) \quad (1)$$

After correcting for baseline drift, the signal was passed through a high-pass filter to eliminate low-frequency noise [33] as it is a standard EEG pre-processing step. Which preserves the brain activity frequency bands, delta-gamma, whilst discarding the more irrelevant slower drifts. The output was the channel's signal ready for artifact removal as in figure 3.

3.2.2 Artifact Removal

In order to cut down on processing time when using the SEED-VII dataset whilst preserving accuracy, 11 channels were extracted out of the 62 channels. The channels were selected based on Dura's [7] results. These channels were FC5, FC1, CZ, P7, PO3, O1, CP2, O2, C4, F4, T8.

After pre-processing, the Blind Source Separation (BSS) artifact removal process begins using Variational Mode-Decomposition - Canonical Correlation Analysis (VMD-CCA) [17]. This method has two parts: Variational Mode Decomposition (VMD), Canonical Correlation Analysis (CCA). VMD non-recursively decomposes the signal into 5 modes using the Hilbert transform. Where each mode contains its own central frequency which is centred around 0 by multiplying the signal with an exponential function that is tuned to estimate the centre frequency of the mode. A mode is a signal in which the count of the local troughs and peaks compared to the number of times it crosses zero differs by no more than one. Afterwards, the bandwidth is estimated using Gaussian smoothness.

Given an observed signal x_0 , which is the noisy version of the true signal x corrupting by noise η

$$x_0 = x + \eta \quad (2)$$

In order to solve this, Tikhonov Regularisation is used to find an optimal solution by balancing 2 terms. The object is to minimise the data fidelity term, which measures how well the solution matches the noisy data, and regularisation term, which penalises large variations in the signal. The solution to the minimisation problem can be obtained in the Fourier domain, giving:

$$\hat{X}(\omega) = \frac{\hat{X}_0(\omega)}{1 + \alpha\omega^2} \quad (3)$$

Where α is a regularisation parameter (we used 400) and ω is the centre frequency. This can be seen as a generalised version of the Weiner Filter [34]. In VMD, the goal is to decompose the signal $x(t)$ into several modes $x_k(t)$, each with a centre frequency ω_k . This is formulated as an optimisation problem. Where the sum of all the modes should equal the original signal:

$$\begin{aligned} \min_{x_k, \omega_k} \sum_{k=1}^K \|\partial_t[(\delta t + j\pi t) * x_k(t)]e^{j\omega_k t}\|_2^2 \\ \text{s.t.} \sum_{k=1}^K x_k(t) = x(t) \end{aligned} \quad (4)$$

This problem is solved by introducing a Lagrangian multiplier and using a penalty term to enforce the constraint that all modes sum to the original signal. In order to solve the constrained optimization problem, VMD uses the Alternating Direction Method of Multipliers (ADMM), a powerful optimization technique. ADMM iteratively updates the modes and their centre frequencies ω_k , alternating between the different parts of the problem to minimize the objective function. The update rule for each mode x_k is given by:

$$x_k^{n+1}(\omega) = \frac{\hat{X}(\omega) - \sum_{i < k} x_i^{n+1}(\omega) - \sum_{i > k} x_i^n(\omega) + \lambda^n(\omega)}{1 + 2\alpha(\omega - \omega_k)^2} \quad (5)$$

As the algorithm continues to iterate, the changes in the modes between iterations become small, as defined by the convergence criterion:

$$\sum_k \left(\frac{\|x_k^{n+1} - x_k^n\|_2^2}{\|x_k^n\|_2^2} \right) < \epsilon \quad (6)$$

Where ϵ is a small tolerance value that controls when the algorithm should stop. The output of VMD separated modes can be visualised in Figure 4. These modes are stacked together to form a matrix X to be used as an input into the CCA section of the algorithm.

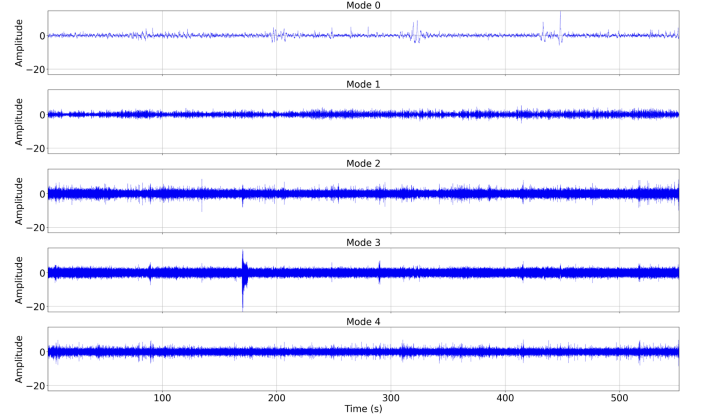


Fig. 4. Motion contaminated data decomposed using VMD into 5 modes. These modes are stacked and used as the input for CCA

The second part of removing the artifacts is using CCA. CCA works by finding the linear combinations of signals from two sets, modes in this case, such that they are maximally correlated. At the end of this pipeline the output is an estimated signal which should be close to artifact free. This method assumes linear, square and stationary mixing. The goal in CCA is to use second order statistics to discover the original independent sources $\hat{S}$ from the mixed signals, modes, X using the linear model: $\hat{S} = WX$ Where we want to compute an un-mixing matrix W by finding linear combinations of the signals so that the two sets are maximally correlated with each other.

Given an input signal x , and y is a linear combination of neighbouring samples $y(t) = x(t-1) + x(t+1)$. The linear combinations of these would be canonical variates:

$$\begin{aligned} x &= \omega_x^T (x - \bar{x}) \\ y &= \omega_y^T (y - \bar{y}) \end{aligned} \quad (7)$$

Where ω_x and ω_y are the weights. CCA tries to find ω_x and ω_y that maximises the correlation ρ between x and y :

$$\rho = \frac{w_x^T C_{xy} w_y}{\sqrt{w_x^T C_{xx} w_x \cdot w_y^T C_{yy} w_y}} \quad (8)$$

Where C_{xx} is the covariance of x , C_{yy} is the covariance of y , and C_{xy} is the cross-covariance between x and y . This is an eigenvalue problem and so to determine the maximum of ρ it is calculated by solving:

$$\begin{aligned} C_{xx}^{-1} C_{xy} C_{yy}^{-1} C_{yx} w_x &= \rho^2 w_x \\ C_{yy}^{-1} C_{yx} C_{xx}^{-1} C_{xy} w_y &= \rho^2 w_y \end{aligned} \quad (9)$$

To construct the un-mixing matrix M , the eigenvectors are sorted by highest correlation first and then stacked to form the un-mixing matrix:

$$W_x = [\omega_{x1}, \omega_{x2}, \dots, \omega_{xm}] \quad (10)$$

Which gives the estimated sources:

$$\hat{S} = W_x X \quad (11)$$

Then to get the clean signal, identify and set the artifact components of $\hat{S}$ to zero. Then reconstruct the clean signal using the inverse of the un-mixing matrix:

$$\hat{X}_{clean} = W^{-1} \hat{S}_{clean} \quad (12)$$

To regain the single channel version after using VMD-CCA, the columns in $\hat{X}_{clean}$ should be added back together. The result of using the VMD-CCA algorithm is shown in Figure 5

Now moving onto the SEED-VII dataset which was used for the emotion classification. An identical processing pipeline and artifact removal technique was used, resulting in Figure 6.

3.3 Feature Extraction

Before performing feature extraction, the EEG signal was segmented into 3 second clips from 120 seconds of data. This was to increase the amount of data given as an input into the CNN. Given the data is recorded at 200 Hz, this will mean that each 3 second clip will contain 600 points of data.

The emotional information used as input to the classifier was captured through a set of non-linear signal features. These are ways to quantify how complex, chaotic or predictable a signal is. The 4 measures we used were: Sample Entropy (SE), Tsallis Entropy (TE), Higuchi Fractal Dimension (HFD), and Hurst Exponent (HE) for each EEG channel.

3.3.1 Sample Entropy

Measures the complexity or unpredictability of the EEG signal. If the signal is very regular, the SE will be low. Whereas, it is more unpredictable, the SE is higher. It is calculated by reviewing short patterns in the data and seeing how often they repeat. SE can be calculated as:

$$SE = -\ln \left(\frac{A_{m+1}(r)}{B_m(r)} \right) \quad (13)$$

Where r is the tolerance for how long to wait before there is no match. Whereas m is the pattern length. A is the number of vectors pairs of length $m+1$. And B is the number of vector pairs of length m .

3.3.2 Tsallis Entropy

Measures the long term-physical behaviour of the system. TE is able to differentiate EEG spikes, bursts or rhythmic patterns in the brain's activity. TE can be calculated as:

$$TE = \left(\frac{1 - \sum_{i=1}^W p_i^q}{q - 1} \right) \quad (14)$$

This is where q is a parameter that controls how much weight rare compared to common events. In this project we used ($q=2$). p_i describes the discrete set of probabilities with W microscopic configurations.

3.3.3 Higuchi Fractal Dimension

Measures the fractal dimension, the roughness, of the EEG signal. Important as different emotions have different levels of roughness. HFD is calculated using Higuchi's algorithm, which splits the signal into sub-sequences and measures how each one fluctuates. For an EEG signal $x = \{x(1), x(2), \dots, x(N)\}$ we construct k new time-series as follows:

$$x_m^k = \{x(m), x(m+k), x(m+2k), \dots\} \quad \text{for } m = 1, 2, \dots, k \quad (15)$$

Each of these k sub-series corresponds to a different starting point m and interval k . The length of each time-series $L_m(k)$ is computed as:

$$L_m(k) = \frac{(N1) \sum_{i=1}^{\frac{N-m}{k}} x(m+ik)x(m+(i1)k)}{k \frac{N-m}{k}} \quad (16)$$

Next, an array of mean lengths $L(k)$ over all m values is calculated we used $k=10$:

$$L(k) = \frac{1}{k} \sum_{m=1}^k L_m(k) \quad (17)$$

Finally, the HFD is obtained from this array by computing the slope:

$$HFD = \frac{\log(L(k))}{\log(1/k)} \quad (18)$$

3.3.4 Hurst Exponent

Measures how predictable the signal is over time. Where $H = 0.5$ means the signal is random (like a coin flip), $H > 0.5$ means the signal long-term positive correlation. $H < 0.5$ means the signal tends to switch directions, meaning more chaotic behaviour. This is helpful as some emotional states may be more predictable (calm, relaxed) or chaotic (fear, anger). HE can be computed as:

$$H = \frac{\log(r/s)}{\log(t)} \quad (19)$$

Where t is the duration of the segment of data and r/s is the corresponding value of the rescaled range of the time series. r is the range of the cumulative deviation from the mean whereas, s is the standard deviation of the time series.

3.4 Deep Learning Model: Convolutional Neural Network

In this project, a CNN is employed as the architecture for emotion classification. Our CNN architecture consists of 3 types of layers: convolutional layers, pooling layers, and fully connected layers. These are preceded by an input layer and usually followed by an output layer for classification. The convolutional layers act as feature extractors by learning spatial hierarchies of patterns through the application of learnable kernels that convolve across the input volume. These kernels slide across the input whilst performing a matrix multiplication to generate a convolved feature map, preserving spatial relationships in the data.

Following the convolutional layers, activation functions such as the Rectified Linear Unit (ReLU) are applied to introduce non-linearity into the network, enabling the model

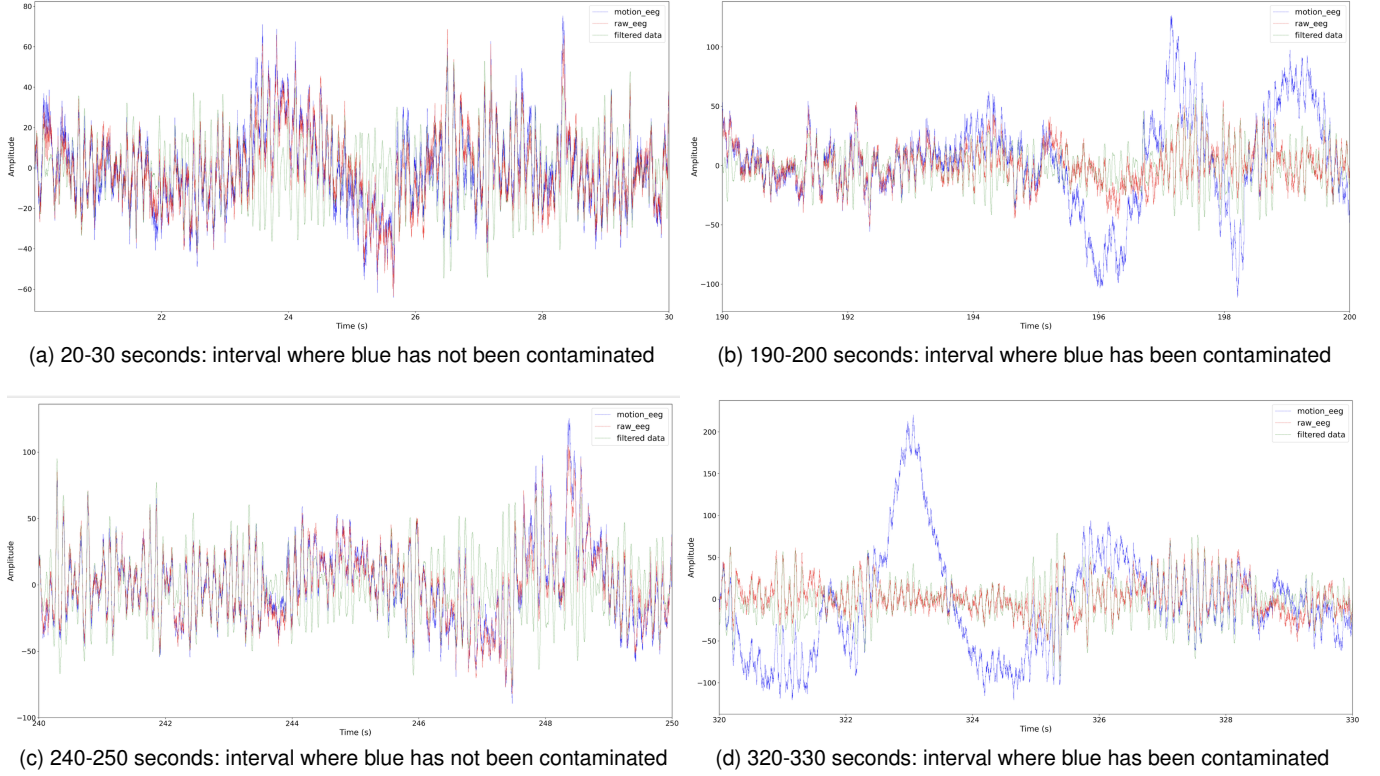


Fig. 5. Zoomed into motion artifact signal at 4 different 10 second intervals. Red is the ground truth signal, blue is the motion contaminated signal, green is the estimated signal which is outputted

TABLE 2

The values or types of the hyper-parameters of our CNN model

Model Hyper-Parameter	Value
Learning rate	0.0023
Weight decay	0.00022
Factor	0.01
Patience	3
Batch Size	128
Pooling method	Maximum Pooling
Activation function	ReLU
Optimisation algorithm	Adam
Number of epochs	30

to capture complex and non-linear patterns. After several rounds of convolutional and pooling operations, the extracted features are passed to one or more fully connected layers. These layers serve as traditional multilayer perceptrons that perform the final classification by mapping the learned high-level features to output classes. The activation function in the final layer produces a probability distribution over the target labels. In our proposed approach, the CNN takes as input the 4 feature vectors for each channel, producing 11x4 EEG-based images. The CNN processes them through a sequence of 3 convolutional-pooling blocks, and finally maps the extracted features to emotion classes using fully connected layers. The design of our CNN model is shown in Figure 7 and the specific hyperparameter values are displayed in Table 2.

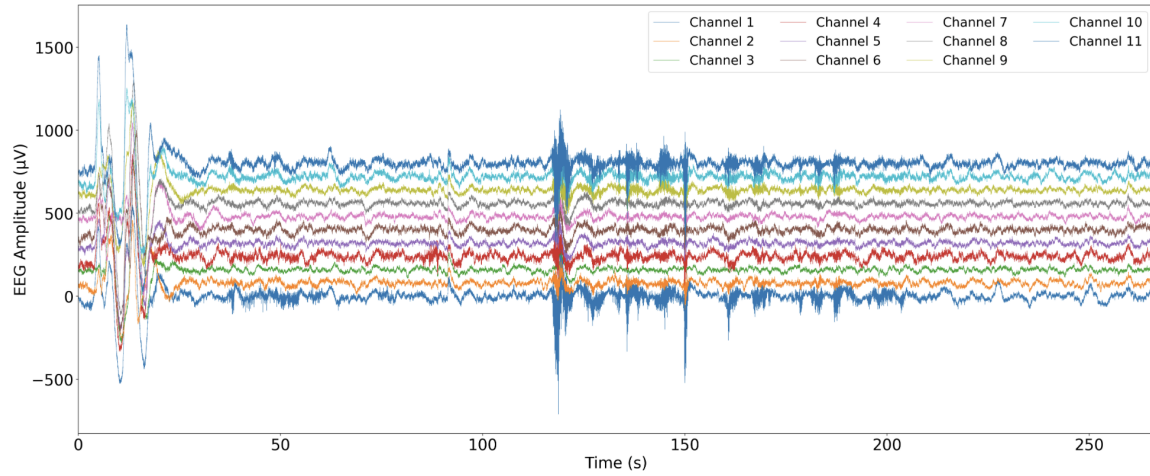
3.5 Validation and Testing

To validate the artifact removal technique, the motion artifact dataset was used, which included a known ground truth signal. The effectiveness of artifact removal was quantitatively assessed using Signal-to-Noise Ratio (SNR) and correlation with the ground truth. SNR was computed by comparing the power of the clean signal to the power of the noise.

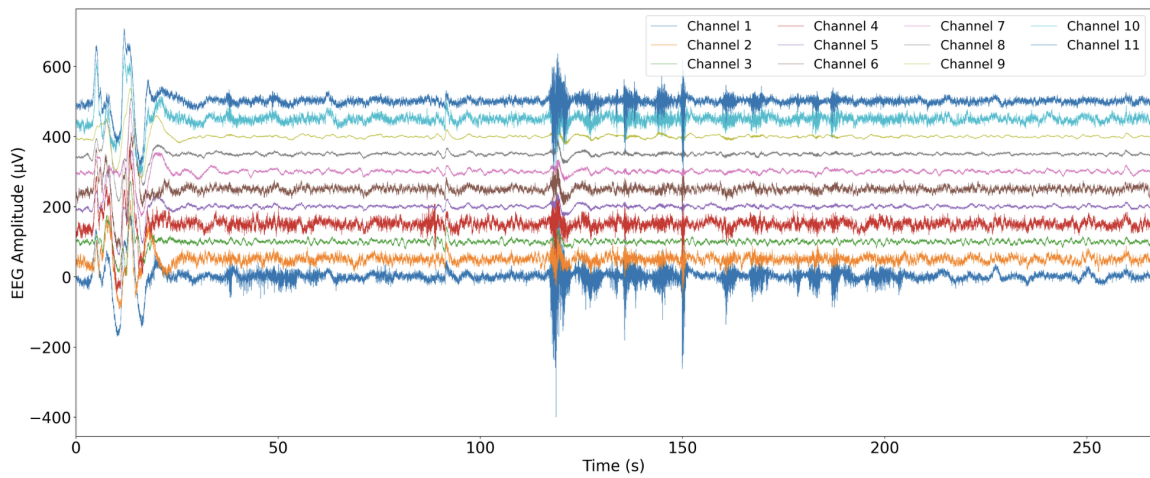
For the emotion classification model, the SEED-VII dataset was partitioned into training (80%), testing (20%) sets. The splits were performed such that no data from the same session or participant overlapped across these sets, thus minimizing the risk of information leakage and overfitting. This ensured that the model was evaluated on entirely unseen data during testing phases.

During model training, the testing set was used to monitor learning progress and guide hyperparameter tuning. Key metrics such as accuracy and loss were computed on the testing set at each epoch. Hyperparameter optimisation of the CNN model was achieved using Ray Tune Bayesian Optimisation [35] to target the highest classification accuracy.

Once the model was trained, final evaluation was carried out on the test set. The model's performance was assessed using several classification metrics, including accuracy, precision, recall, F1-score, and confusion matrices. These metrics were computed for both the overall model and across individual emotion classes to provide a detailed view of classifier behaviour.



(a) SEED-VII RAW EEG



(b) SEED-VII after VMD-CCA EEG

Fig. 6. Comparison of before VMD-CCA was applied (up) and after (down). The amplitude of artifacts have been significantly reduced

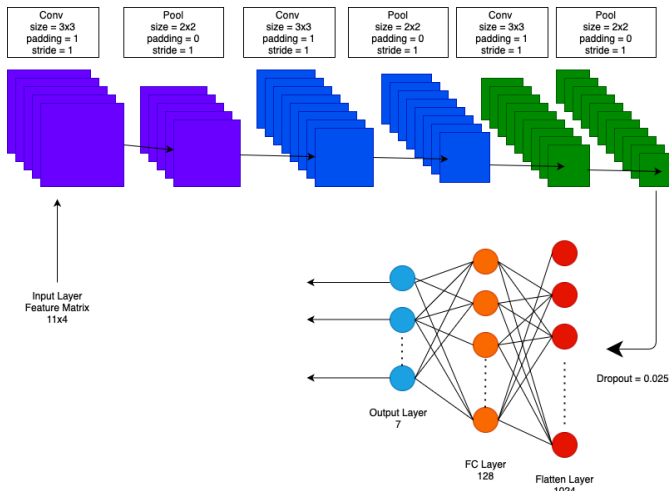


Fig. 7. Overview of the CNN architecture used. The network receives input of size (11x4) and processes it through a sequence of convolutional and pooling layers. The final fully connected layer outputs probabilities across emotion classes

4 RESULTS

This section evaluates the success of the artifact removal method by using Signal to Noise Ratio (SNR) using the

motion artifact dataset. To evaluate how well the artifact removal method works in a emotion classification setting: classification accuracy, precision, recall, F1-score, confusion matrix, and the training/testing losses are used.

4.1 Artifact Removal

Using the motion contaminated dataset, the SNR before removing the artifacts was -5.034. Whereas, after removing the artifacts the SNR jumped up to 2.969. This displays that the techniques were extremely successful.

4.2 Model Performance

The dataset was preprocessed by removing any data with ratings ≤ 0.5 , ensuring that the CNN model only learned from data that accurately reflected the intended emotional response. This process resulted in a class imbalance, as shown in Figure 8. After the data was preprocessed, a total of 53,240 feature matrices were available for the model. These were split into 42,592 training examples and 10,648 testing examples. Each EEG segment was represented by 4 feature vectors (e.g., entropy measures), resulting in 212,960 total vectors across all examples.

Due to the removal of some data based on participants' low post-stimulus ratings, a slight class imbalance was introduced, as shown in Figure 8. This highlights the inherent difficulty in eliciting no emotional response, with the neutral class having marginally fewer instances than others. Despite this, the confusion matrix in Figure 9 demonstrates strong model performance, with clear diagonal dominance indicating high classification accuracy. Misclassifications are minimal, with off-diagonal values remaining $\leq 2\%$. The greatest confusion is observed in the neutral class, likely due to its inherently subtle and ambiguous nature.

Further insight is provided by the class-wise evaluation metrics in Table 3 shows consistently high F1-scores across all emotion categories, ranging from 0.91 to 0.93. This suggests the model is well-balanced in its ability to detect true positives while minimizing both false positives and false negatives. Notably, Anger and Surprise achieved the highest F1-scores (0.93), suggesting that the model effectively captures features specific to these emotions. The lowest precision was observed for Disgust (0.91), indicating that there is some confusion with Anger and Sad. However, even in this case, the recall remained high at 0.92, further supporting the model's robustness.

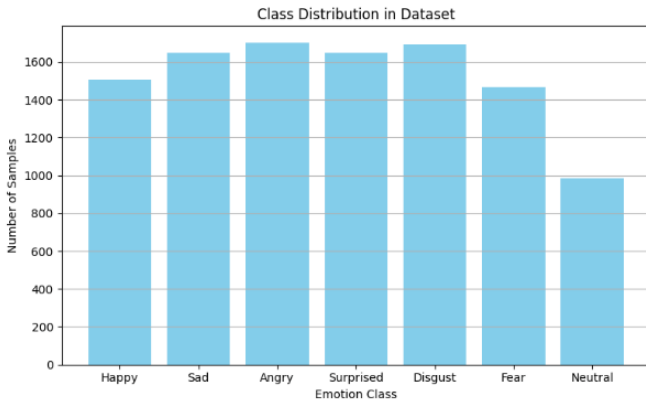


Fig. 8. Distribution of samples across classes. The x-axis represents the emotion classes, and the y-axis shows the number of samples in each class. This highlights the imbalance with the neutral class in the dataset, which may influence model performance

Emotion	Precision	Recall	F1-score
Happy	0.92	0.91	0.92
Surprise	0.93	0.93	0.93
Disgust	0.91	0.92	0.91
Fear	0.93	0.92	0.92
Anger	0.93	0.93	0.93
Sad	0.91	0.91	0.91
Neutral	0.91	0.90	0.91

TABLE 3
Classification metrics for each emotion class.

4.3 Training Metrics

The accuracy of the model is shown in Figure 10. Due to the small dataset size, there are minor fluctuations in the loss function. However, the overall trend demonstrates consistent learning. By the end of training, the model achieves a stable accuracy of 91.5%.

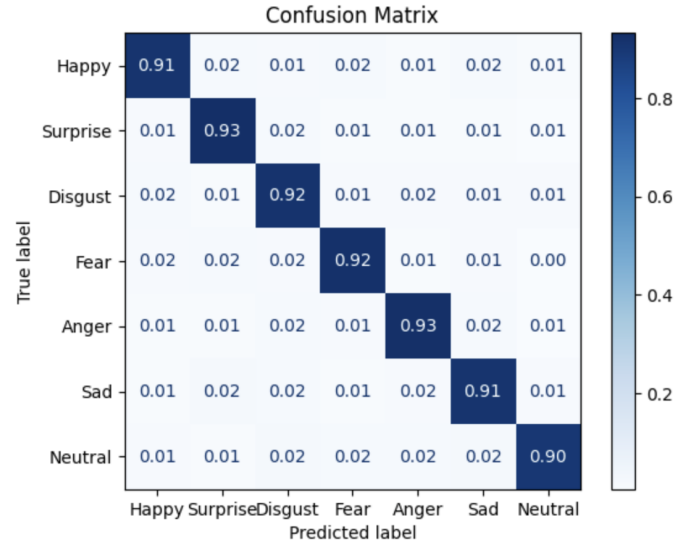


Fig. 9. Normalized confusion matrix showing the percentage of predictions made by the CNN-based emotion classification model. The y-axis represents the true emotion classes, and the x-axis shows the predicted classes. Each cell indicates the proportion of predictions for a given true class. Diagonal values reflect correct classifications, while off-diagonal entries show where the model tends to confuse similar emotions.

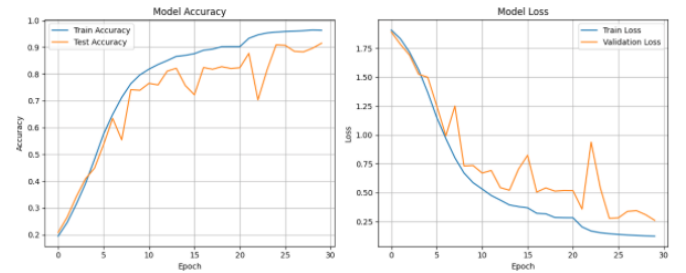


Fig. 10. Training and validation accuracy and loss curves over 30 epochs for the CNN-based emotion classification model

4.4 Deliverables

Nearly all deliverables were successfully achieved. Pre-processing techniques for artifact and noise removal were effectively implemented. Feature extraction pipelines were also developed to structure the EEG signals into meaningful representations suitable for deep learning. A convolutional neural network (CNN) model was implemented and extensively fine-tuned, achieving a classification accuracy exceeding 90%, thereby fulfilling the performance target set for the project. This strong performance rendered the LSTM implementation unnecessary, as the CNN alone was sufficient to meet the classification objectives with high accuracy and reliability. Its ability to generalize well across emotion classes validates both the design of the model and the quality of the data processing pipeline. Although we did not fully achieve the deliverable of producing a polished software package, we consider this secondary to the core functional goals of the project. Given that the codebase is modular and the functionality is complete, the creation of a refined software package could be completed after the main objectives of this dissertation have been addressed.

5 EVALUATION

This section evaluates the overall effectiveness of the CNN-based EEG emotion recognition system developed in this project. It critically assesses the extent to which the project objectives were met, the quality of the results obtained, and the effectiveness of key design decisions — including artifact removal, data preprocessing, model architecture, and training strategy. The strengths and limitations of the system are discussed, alongside reflections on the software engineering practices and project management approaches employed throughout development.

5.1 Effectiveness of Artifact Removal Method

The artifact removal method proved to be highly effective across both synthetic (motion-contaminated) and real-world (SEED-VII) EEG datasets without requiring manual tuning or subject-specific knowledge. This generalisability is a major strength, suggesting that the method can be reliably applied in practical rehabilitation settings without domain expertise. The SNR increased significantly after applying the method, and this improvement was strongly correlated with high classification performance. Notably, the CNN achieved over 91% accuracy in emotion classification on the cleaned signals — a level of performance unlikely to be reached if the artifact removal method had inadvertently distorted relevant EEG features. This demonstrates that the method successfully reduces motion-induced noise while preserving emotion-related signal integrity.

However, the approach does present some limitations. The processing time required for applying the algorithm, particularly VMD in the VMD-CCA, is relatively high, making real-time use challenging. A potential solution for real-time deployment would be to process shorter data segments, such as two-second windows, to reduce latency. Additionally, in a very small number of cases — 3 out of 1,600 trials (less than 0.2%) — the VMD algorithm failed to converge, resulting in those trials being discarded. The cause of this failure remains unclear, though its rarity suggests that it does not significantly affect the robustness of the method. Nonetheless, understanding and addressing these convergence failures would further improve the reliability of the system in clinical applications.

5.2 Strengths of the System

The pipeline exhibits strong generalisability, making it applicable across diverse patients and clinical settings without requiring domain-specific expertise. For instance, the artifact removal stage leverages a Blind Source Separation (BSS) technique, which does not require prior knowledge of the specific source of interference — such as eye movements or limb motion — to effectively isolate and remove noise. This allows hospital staff to deploy the system with minimal manual intervention, enabling them to focus more on patient care and rehabilitation. Additionally, the feature extraction process is entirely automated, requiring no user input. The only component that may benefit from dataset-specific tuning is the CNN model; however, this can be addressed in future work by retraining or fine-tuning the model using additional data sources.

Another significant strength of this project is its modular architecture. Each stage of the pipeline — from preprocessing to classification — is independently replaceable or extendable. This facilitates future enhancements, such as the integration of improved artifact removal methods, alternative feature extraction techniques, or more sophisticated classification models, without requiring a complete system redesign.

The CNN's performance remained robust even with class imbalance, and training adjustments such as batch size scaling and learning rate decay led to smoother convergence. Notably, the model outperformed existing approaches in the literature: Taran's pipeline which achieved 88% [27] with only 4 emotions and Zhong's graph neural network reached 85% accuracy [10] without introducing artifacts and therefore artifact removal techniques. The ability to classify emotional responses with high accuracy is particularly valuable in clinical rehabilitation, where accurate assessment of patient affect can significantly influence treatment decisions and recovery outcomes.

5.3 Limitations and Challenges

A notable limitation of the model is its reliance on the subjective self-reported ratings to determine how well an emotion was simulated. Although it improved training accuracy, this may have inadvertently prevented some generalisability with the model as it may not learn the edge cases of each emotion. The balance must be struck between only inputting correct emotion data per label and inputting a variety of cases per emotion. However, to fully understand the effects of this future work must perform validation with alternative datasets.

Furthermore, the removal of data with low ratings, while improving label quality, led to a slight class imbalance — particularly a reduced number of neutral class examples. This imbalance may affect generalisation across emotional categories. However, this limitation is expected to have minimal impact on overall performance, as the neutral emotion is inherently more ambiguous and less distinct in EEG signal patterns compared to strongly valenced emotions such as happiness or sadness. Moreover, the model demonstrated strong classification performance across all classes, including neutral, as evidenced by the confusion matrix. This suggests that the remaining neutral samples still contained sufficiently representative signal patterns for the model to learn meaningful distinctions.

This is particularly important in the context of the project's real-world aim: supporting healthcare professionals in tailoring rehabilitation plans by monitoring patients' emotional responses during active therapy. In such settings, it is especially important to distinguish between positive engagement and negative emotional responses, as these are the key indicators of whether an exercise is both challenging and tolerable. While misclassifying neutral states may have a minor impact, accurately identifying distress or discomfort is far more critical. The system's robustness in detecting emotional extremes ensures that it remains a valuable tool in clinical environments, even with a slightly under-represented neutral class. In future work, integrating class-balancing techniques or capturing more balanced real-world data could further enhance its practical reliability.

5.4 Software Engineering Processes

At the outset of the project, a Gantt chart was used to plan the timeline and allocate resources across the major stages of development. While this provided an initial structure, the iterative nature of the work revealed that certain stages — particularly preprocessing and feature extraction — required significantly more time and attention than originally anticipated. These stages proved to be more critical to the overall system performance than initially assumed, and as such, the deep learning component received comparatively less development time. In retrospect, the project plan should have been revised dynamically to reflect these evolving priorities, allowing for better time management and resource allocation.

Despite this, the core software pipeline — from raw EEG input to final emotion classification — remained largely consistent with the original design. This stability reflects sound early architectural decisions and a clear alignment with the project’s research question. The modular structure of the pipeline also enabled testing at multiple stages, which was consistently used to verify the correctness of individual components such as artifact removal, feature extraction, and data filtering.

However, the software engineering process could have been strengthened through more comprehensive background research and method selection in the initial stages. More time invested in evaluating different algorithms and tools upfront would likely have reduced implementation overhead later and potentially freed up time to develop a more polished, user-friendly software package. Although the system is functionally complete and the codebase is modular, the lack of a finished software product highlights the trade-off between rigorous method development and final presentation.

5.5 Future Work

Although this project successfully demonstrated the feasibility of EEG-based emotion recognition in active rehabilitation scenarios, several avenues remain for future development.

First, transitioning from an offline pipeline to a real-time system is a key next step. In real-world rehabilitation settings, continuous emotion monitoring is necessary to provide immediate feedback to therapists and adjust treatment plans dynamically. Achieving real-time capability will require optimising the computational efficiency of the artifact removal and feature extraction stages. One proposed solution is to process shorter EEG segments — such as two-second windows — to reduce latency and ensure smooth streaming performance. This will require further testing and algorithmic refinement to maintain classification accuracy under tight time constraints.

Second, deploying and validating the system in uncontrolled, real-world environments — such as rehabilitation clinics or hospital wards — is essential. Real-world trials would allow for assessing the system’s robustness against unpredictable noise sources and participant behaviours not present in curated datasets. This would also provide an opportunity to gather more balanced and diverse training data, particularly for under-represented emotional states like neutrality.

Third, future versions of the model could incorporate adaptive mechanisms that allow it to fine-tune itself over time based on individual patient data. Personalisation could improve classification accuracy by accounting for subject-specific EEG patterns without requiring manual calibration.

Finally, extending the pipeline to work with wearable or portable EEG devices could also support home-based or outpatient monitoring, broadening the reach and accessibility of emotionally intelligent rehabilitation support.

6 CONCLUSION

This project aimed to remove motion-induced artifacts from EEG signals to accurately detect emotions to ultimately assess patients’ emotions during active rehabilitation. This, in turn, helps healthcare professionals determine the most appropriate course of rehabilitation to reduce distress and optimise recovery outcomes. This project successfully developed a robust and generalisable EEG-based emotion recognition pipeline capable of classifying patient emotions during active rehabilitation. The findings demonstrate high classification accuracy despite motion-induced artifacts, with the accuracy at 92%. This supports the hypothesis that, with appropriate artifact removal and deep learning techniques, reliable emotion classification is possible even in noisy, real-world EEG data. By integrating automated artifact removal using Blind Source Separation, fully unsupervised feature extraction, and a deep learning-based classification model, the pipeline reduces reliance on domain expertise. This makes the approach suitable for clinical environments where usability, consistency, and minimal manual intervention are critical.

A key contribution of this research is that the model outperformed previously published approaches, demonstrating over 90% classification accuracy even while supporting more emotion classes and using real-world noisy data. Additionally, the modularity of the pipeline, which allows future researchers or developers to easily substitute or improve individual components which makes the system highly adaptable and extendable for future innovations. This highlights the potential of EEG-based systems for objective emotional assessment, particularly in patient rehabilitation where traditional facial or self-report methods are unreliable.

The practical implications of this work are significant: accurate emotional assessment can support more personalised rehabilitation, helping clinicians understand how different exercises impact emotional states and, by extension, recovery outcomes. The pipeline could be used in future hospital settings to monitor patient emotions in real time and optimise rehabilitation strategies accordingly.

One limitation of this project is the reliance on self-reported emotion ratings, which introduces a degree of subjectivity into the dataset. While these ratings provide valuable insight into participants’ perceived emotional responses, they are inherently influenced by individual interpretation, mood, and other contextual factors that may not be consistent across sessions or participants. As a result, the ground truth labels used to train and evaluate the model may not fully represent objective emotional states. To address this, future work could focus on validating the model

using alternative datasets or real-world data collected in more naturalistic settings or through physiological markers with less reliance on self-assessment. Additionally, conducting longitudinal studies or real-time trials could provide a more robust understanding of how the model performs in practical applications, such as in rehabilitation or affective computing systems.

Further work could focus on developing methods to enable real-time application of the model, allowing for immediate feedback to healthcare professionals during patient interaction or monitoring. This would enhance the practical utility of the system, particularly in clinical or rehabilitative contexts where timely insights are critical. Additionally, deploying the model in a real-world environment and validating its performance in a clinical setting would provide a more rigorous assessment of its reliability, generalisability, and effectiveness in practice. Such trials would help identify potential challenges, ultimately guiding future improvements in both model design and deployment strategy.

REFERENCES

- [1] Ekman, Paul. "Are there basic emotions?." (1992): 550.
- [2] Trudel-Fitzgerald, C., Qureshi, F., Appleton, A.A. and Kubzansky, L.D., 2017. A healthy mix of emotions: underlying biological pathways linking emotions to physical health. *Current Opinion in Behavioral Sciences*, 15, pp.16-21.
- [3] Ostir, G.V., Berges, I., Ottenbacher, M., Graham, J.E. and Ottenbacher, K.J., 2008. Positive emotion following a stroke. *Journal of rehabilitation medicine: official journal of the UEMS European Board of Physical and Rehabilitation Medicine*, 40(6), p.477.
- [4] Seale, G.S., Berges, I.M., Ottenbacher, K.J. and Ostir, G.V., 2010. Change in positive emotion and recovery of functional status following stroke. *Rehabilitation psychology*, 55(1), p.33.
- [5] Eslinger, P.J., Parkinson, K. and Shamay, S.G., 2002. Empathy and social-emotional factors in recovery from stroke. *Current opinion in neurology*, 15(1), pp.91-97.
- [6] Teplan, M., 2002. *Fundamentals of EEG measurement. Measurement science review*, 2(2), pp.1-11.
- [7] Dura, A. and Wosiak, A., 2021. EEG channel selection strategy for deep learning in emotion recognition. *Procedia Computer Science*, 192, pp.2789-2796.
- [8] Yildirim, E., Kaya, Y. and Kiliç, F., 2021. A channel selection method for emotion recognition from EEG based on swarm-intelligence algorithms. *IEEE Access*, 9, pp.109889-109902.
- [9] Lin, D., Zhang, J., Li, J., Calhoun, V.D., Deng, H.W. and Wang, Y.P., 2013. Group sparse canonical correlation analysis for genomic data integration. *BMC bioinformatics*, 14, pp.1-16.
- [10] Zhong, P., Wang, D. and Miao, C., 2020. EEG-based emotion recognition using regularized graph neural networks. *IEEE Transactions on Affective Computing*, 13(3), pp.1290-1301.
- [11] Chen, X., Xu, X., Liu, A., Lee, S., Chen, X., Zhang, X., McKeown, M.J. and Wang, Z.J., 2019. Removal of muscle artifacts from the EEG: A review and recommendations. *IEEE Sensors Journal*, 19(14), pp.5353-5368.
- [12] Mateo, J., Torres, A.M., Soria, C. and Santos, J.L., 2013. A method for removing noise from continuous brain signal recordings. *Computers & Electrical Engineering*, 39(5), pp.1561-1570.
- [13] Hassani, M. and Karami, M.R., 2015. Noise estimation in electroencephalogram signal by using volterra series coefficients. *Journal of Medical Signals & Sensors*, 5(3), pp.192-200.
- [14] Winkler, I., Haufe, S. and Tangermann, M., 2011. Automatic classification of artifactual ICA-components for artifact removal in EEG signals. *Behavioral and brain functions*, 7, pp.1-15.
- [15] Lin, S.L., 2021. The Application of machine learning ICA-VMD in an intelligent diagnosis system in a low SNR environment. *Sensors*, 21(24), p.8344.
- [16] Zeng, K., Chen, D., Ouyang, G., Wang, L., Liu, X. and Li, X., 2015. An EEMD-ICA approach to enhancing artifact rejection for noisy multivariate neural data. *IEEE transactions on neural systems and rehabilitation engineering*, 24(6), pp.630-638.
- [17] Hossain, M.S., Reaz, M.B.I., Chowdhury, M.E., Ali, S.H., Bakar, A.A.A., Kiranyaz, S., Khandakar, A., Alhatou, M. and Habib, R., 2022. Motion artifacts correction from EEG and fNIRS signals using novel multiresolution analysis. *IEEE Access*, 10, pp.29760-29777.
- [18] Chen, Q., Li, Y. and Yuan, X., 2021. A hybrid method for muscle artifact removal from EEG signals. *Journal of Neuroscience Methods*, 353, p.109104.
- [19] Chen, X., Chen, Q., Zhang, Y. and Wang, Z.J., 2018. A novel EEMD-CCA approach to removing muscle artifacts for pervasive EEG. *IEEE Sensors Journal*, 19(19), pp.8420-8431.
- [20] Murugappan, M., Ramachandran, N. and Sazali, Y., 2010. Classification of human emotion from EEG using discrete wavelet transform. *Journal of biomedical science and engineering*, 3(04), p.390.
- [21] Krishna, A.H., Sri, A.B., Priyanka, K.Y.V.S., Taran, S. and Bajaj, V., 2019. Emotion classification using EEG signals based on tunable-Q wavelet transform. *IET Science, Measurement & Technology*, 13(3), pp.375-380.
- [22] Slimen, I.B., Boubchir, L., Mbarki, Z. and Seddik, H., 2020. EEG epileptic seizure detection and classification based on dual-tree complex wavelet transform and machine learning algorithms. *Journal of biomedical research*, 34(3), p.151.
- [23] Hazarika, N., Chen, J.Z., Tsoi, A.C. and Sergejew, A., 1997. Classification of EEG signals using the wavelet transform. *Signal processing*, 59(1), pp.61-72.
- [24] Alazrai, R., Homoud, R., Alwanni, H. and Daoud, M.I., 2018. EEG-based emotion recognition using quadratic time-frequency distribution. *Sensors*, 18(8), p.2739.

- [25] Khare, S.K., Nishad, A., Upadhyay, A. and Bajaj, V., 2020. Classification of emotions from EEG signals using time-order representation based on the S-transform and convolutional neural network. *Electronics Letters*, 56(25), pp.1359-1361.
- [26] Islam, M.R., Islam, M.M., Rahman, M.M., Mondal, C., Singha, S.K., Ahmad, M., Awal, A., Islam, M.S. and Moni, M.A., 2021. EEG channel correlation based model for emotion recognition. *Computers in Biology and Medicine*, 136, p.104757.
- [27] Taran, S. and Bajaj, V., 2019. Emotion recognition from single-channel EEG signals using a two-stage correlation and instantaneous frequency-based filtering method. *Computer methods and programs in biomedicine*, 173, pp.157-165.
- [28] Masood, K. and Alghamdi, M.A., 2019. Modeling mental stress using a deep learning framework. *IEEE access*, 7, pp.68446-68454.
- [29] Alzubaidi, L., Zhang, J., Humaidi, A.J., Al-Dujaili, A., Duan, Y., Al-Shamma, O., Santamaría, J., Fadhel, M.A., Al-Amidie, M. and Farhan, L., 2021. Review of deep learning: concepts, CNN architectures, challenges, applications, future directions. *Journal of big Data*, 8, pp.1-74.
- [30] Sweeney KT, Ayaz H, Ward TE, Izzetoglu M, McLoone SF, Onaral B. A Methodology for Validating Artifact Removal Techniques for Physiological Signals. *IEEE Trans Info Tech Biomed* 16(5):918-926; 2012 (Sept).
- [31] Wei-Bang Jiang, Xuan-Hao Liu, Wei-Long Zheng and Bao-Liang Lu, SEED-VII: A Multimodal Dataset of Six Basic Emotions with Continuous Labels for Emotion Recognition, *IEEE Transactions on Affective Computing*, 2024.
- [32] De Cheveigné, A. and Arzounian, D., 2018. Robust detrending, rereferencing, outlier detection, and inpainting for multichannel data. *NeuroImage*, 172, pp.903-912.
- [33] Winkler, I., Debener, S., Müller, K.R. and Tangermann, M., 2015, August. On the influence of high-pass filtering on ICA-based artifact reduction in EEG-ERP. In *2015 37th annual international conference of the IEEE engineering in medicine and biology society (EMBC)* (pp. 4101-4105). IEEE.
- [34] Dragomiretskiy, K. and Zosso, D., 2013. Variational mode decomposition. *IEEE transactions on signal processing*, 62(3), pp.531-544.
- [35] Running tune experiments with bayesopt (no date) Running Tune experiments with BayesOpt - Ray 2.44.1. Available at: https://docs.ray.io/en/latest/tune/examples/bayesopt_example.html